JANANI D

AI Engineer | Machine Learning | Agentic AI & LLMs | Python | FastAPI, React & Node.js

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PROFESSIONAL SUMMARY

AI Engineer with 2 years of experience building production AI systems for logistics, fleet management and enterprise clients. Owns solutions end to end — from problem understanding and database design to agentic AI pipelines, FastAPI backends and React frontends. Experienced in LangGraph multi-agent systems, LLM tool-calling architecture, and clustering and route-optimization algorithms that replace manual planning with automated, explainable decision systems.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, HTML5, CSS3

AI / ML: LangGraph, LangChain, Agentic AI, LLM Tool-Calling, Multi-Agent Systems, RAG, NLP, Computer Vision, scikit-learn, pandas, NumPy

Optimization: K-Means, DBSCAN, Haversine Distance, Google OR-Tools (Bin Packing), Capacity Planning, Route Optimization, Time-Series Forecasting

Backend: FastAPI, Django, Django REST Framework, Frappe, Node.js, Socket.IO, REST API Design

Frontend: React.js, Next.js, Vite, Leaflet.js, HTML5, CSS3

Databases: PostgreSQL, MySQL, MariaDB, MongoDB, Redis

Cloud & Tools: Docker, Google Cloud Platform, Git, Postman, Power BI, Google Maps API, n8n

PROFESSIONAL EXPERIENCE

AI Engineer — Ko Innovation Software Solutions

Tiruchirappalli, Tamil Nadu | *November 2024 – Present*

AnrilX — Agentic AI Platform for Enterprise ERP | *Python, FastAPI, LangGraph, PostgreSQL, Redis, Next.js, Docker*

- Built a single-tenant agentic AI platform that answers business questions against live enterprise SAP data through a router agent that delegates to specialized module-expert agents.
- Designed a secure gateway as the sole write path to the ERP system, enforcing policy-based access checks, short-lived scoped authentication tokens, human-in-the-loop approval workflows and full audit logging on every write operation.
- Implemented a multi-LLM model-routing system with ranked candidate models per role, automatic failover on timeout, rate-limit and error conditions, and live reordering based on cost, latency and error rate.
- Developed a no-code automation engine supporting email- and schedule-triggered workflows with agent execution, document generation and multi-channel notifications (Slack, Teams, WhatsApp, email).

ITP — Intelligent Transport Planner, KoFleetz Fleet Platform | *Frappe (Python), React, Vite, LangGraph, Socket.IO, Leaflet.js, MariaDB*

- Engineered an AI-driven transport planner that automates vehicle and driver assignment for daily bookings through a deterministic multi-stage pipeline covering validation, consolidation, capacity split, availability check, vehicle scoring, conflict-free allocation and SLA verification.
- Designed a conversational AI assistant exposing 7+ domain tools (run plan, explain assignment, what-if simulation, reassign, approve) with LangGraph checkpointed multi-turn conversations and a curated knowledge base for grounded, explainable responses.
- Delivered an end-to-end vendor bidding module for transport procurement, supporting sealed and reverse-auction bidding with explainable match scoring, two-sided negotiation threads and automatic award hand-off into trip and billing workflows.
- Architected a 12+ entity domain model with a one-way dependency design that keeps the core platform untouched, and built two React single-page applications: an internal planner with live Socket.IO updates and Leaflet route visualization, and an external vendor portal with role-based access control.

Machine Learning Intern — NIT Trichy | *Python, scikit-learn, pandas, Tkinter*
Tiruchirappalli, Tamil Nadu | June 2023 – August 2023

- Trained a regression model on historical BigMart sales data to forecast item-level sales from multiple product and outlet features, covering data preparation, feature selection, model training and evaluation.
- Created a Tkinter desktop interface enabling non-technical users to enter parameters and view sales forecasts directly, without touching code.

PROJECTS

Capacity Planning & Fleet Allocation — Dairy Client | *Python, FastAPI, React, Google Maps API*

- Designed a capacity-aware vehicle allocation engine for a dairy client's milk collection network, assigning vans to collection points by live vehicle location and load capacity, then routing each van to the most efficient chilling center within the processing time window required for perishable goods.
- Integrated Google Maps Distance, Directions and Travel Time APIs for route optimization, replacing a fully manual process that kept no record of which van collected from which point; solution approved by the client.

Delivery & Shuttle Route Clustering | *Python, FastAPI, Google OR-Tools, K-Means, DBSCAN, Google Maps API*

- Implemented a dynamic stop-clustering engine that derives daily ridership per stop through database filtering by academic year and attendance, then clusters stops by route proximity and vehicle capacity to adapt van allocation to actual daily demand.
- Extended the engine to delivery route planning using Google OR-Tools bin packing with real-time location data, improving vehicle utilization and reducing the number of vehicles required per run.

AI-Based Total Electron Content (TEC) Prediction | *Python, Time-Series Analysis, Regression*

- Developed a machine learning model to predict Total Electron Content variations during solar flares, storms and earthquakes using real-world IONOLAB and OMNI Web datasets.
- Applied time-series analysis and regression techniques and generated visual plots and trend graphs for scientific interpretation and geospatial analysis.

EDUCATION

B.Tech, Artificial Intelligence and Data Science | CGPA: 9.02 | **Department 2nd Rank Holder**
Saranathan College of Engineering, Tiruchirappalli | 2021 – 2025

Higher Secondary Certificate (HSC), Biology & Mathematics | 94.4%
Sai Vidhyalaya, Central Board | 2020 – 2021

Secondary School Leaving Certificate (SSLC) | 81%
Sai Vidhyalaya, Central Board | 2018 – 2019

CERTIFICATIONS

Data Analytics with Python and Power BI — Grade A, Higher Package Skill Training

ACHIEVEMENTS

- Awarded **Department 2nd Rank** in Artificial Intelligence and Data Science for academic performance across the B.Tech program.
- Received Certificate of Merit from Saranathan College of Engineering.
- Won Third Prize in intra-college paper presentation competition on cryptocurrency.
- Competed in a 48-hour live hackathon conducted by TN Startup, and in a hackathon hosted by PSG College of Technology.